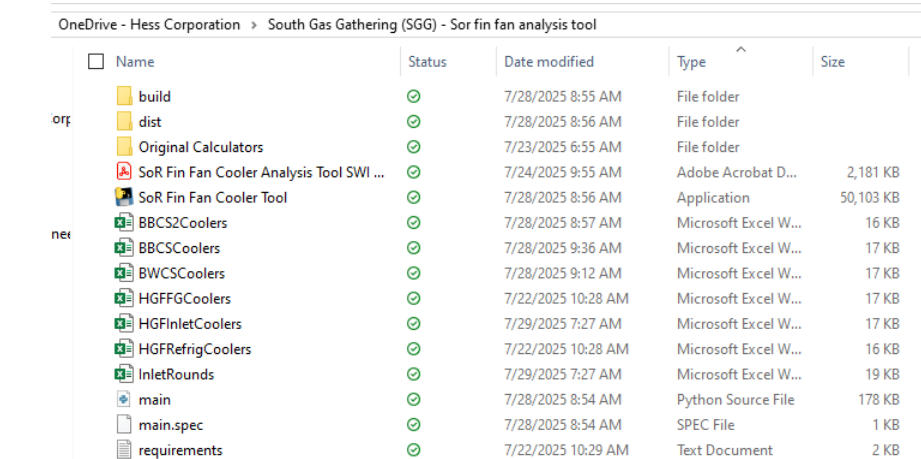


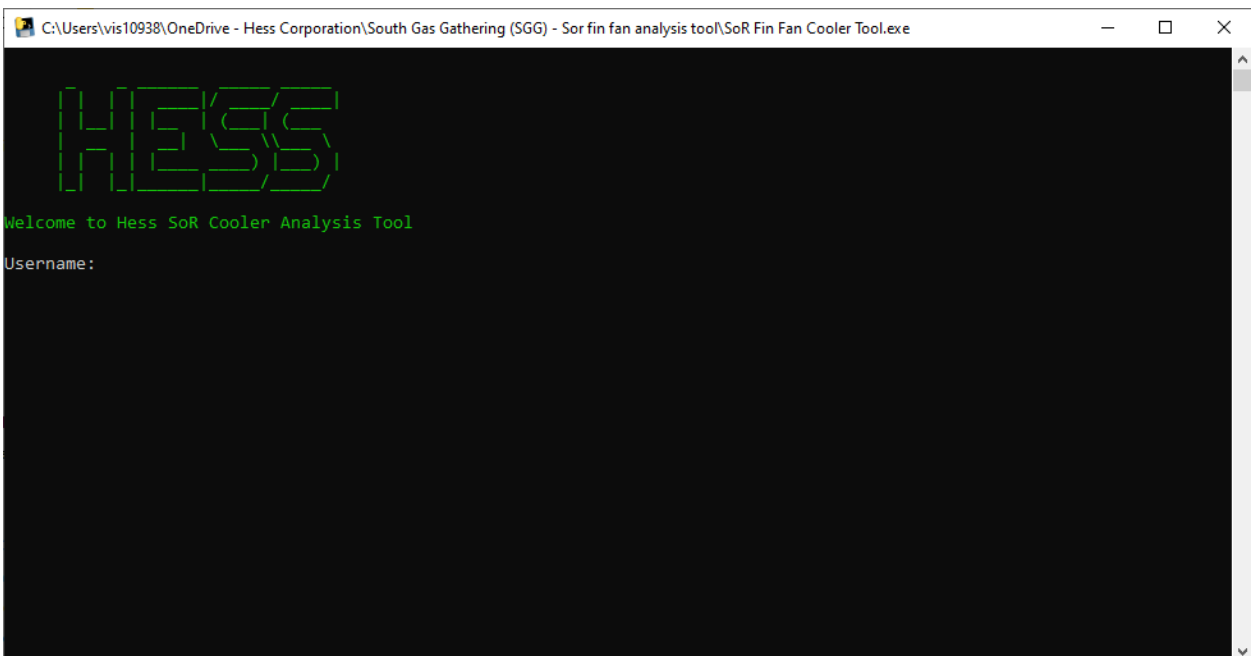
SoR Fin Fan Cooler Analysis Tool – User Guide

1. Ensure that the workspace folder has the application accompanied by the excel files, this pdf, etc. similar to the view below:



OneDrive - Hess Corporation > South Gas Gathering (SGG) - Sor fin fan analysis tool					
☐	Name	Status	Date modified	Type	Size
	build	✓	7/28/2025 8:55 AM	File folder	
	dist	✓	7/28/2025 8:56 AM	File folder	
	Original Calculators	✓	7/23/2025 6:55 AM	File folder	
	SoR Fin Fan Cooler Analysis Tool SWI ...	✓	7/24/2025 9:55 AM	Adobe Acrobat D...	2,181 KB
	SoR Fin Fan Cooler Tool	✓	7/28/2025 8:56 AM	Application	50,103 KB
	BBCS2Coolers	✓	7/28/2025 8:57 AM	Microsoft Excel W...	16 KB
	BBCSCoolers	✓	7/28/2025 9:36 AM	Microsoft Excel W...	17 KB
	BWCSCoolers	✓	7/28/2025 9:12 AM	Microsoft Excel W...	17 KB
	HGFFGCoolers	✓	7/22/2025 10:28 AM	Microsoft Excel W...	17 KB
	HGFinletCoolers	✓	7/29/2025 7:27 AM	Microsoft Excel W...	17 KB
	HGFRfrigCoolers	✓	7/22/2025 10:28 AM	Microsoft Excel W...	16 KB
	InletRounds	✓	7/29/2025 7:27 AM	Microsoft Excel W...	19 KB
	main	✓	7/28/2025 8:54 AM	Python Source File	178 KB
	main.spec	✓	7/28/2025 8:54 AM	SPEC File	1 KB
	requirements	✓	7/22/2025 10:29 AM	Text Document	2 KB

2. Next, access the application titled 'SoR Fin Fan Cooler Tool.exe'. This should be the view:

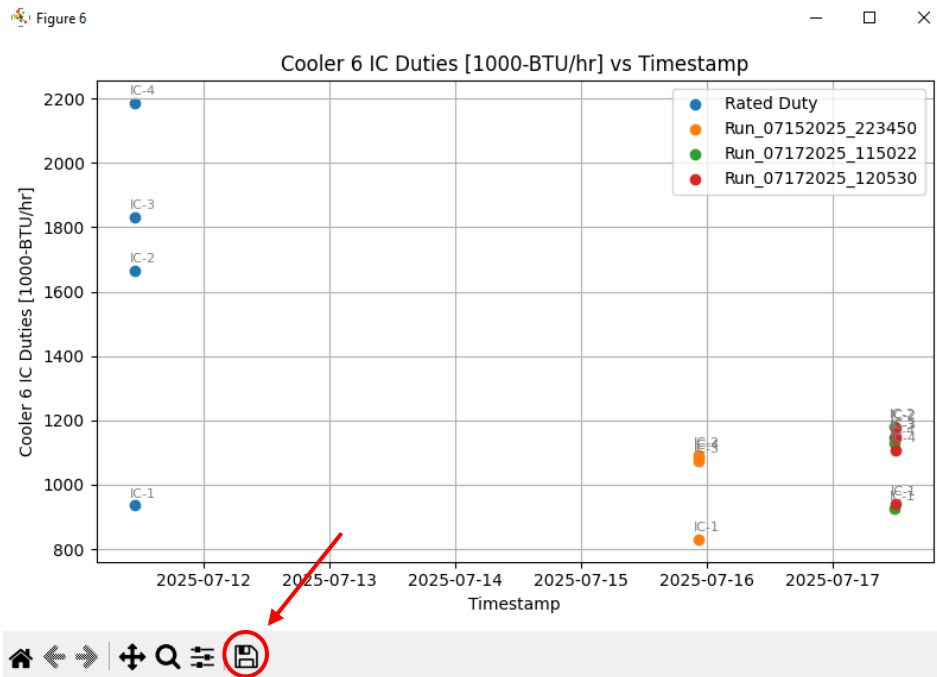


3. Input Hess credentials (Username: first.m.last@hess.com)
4. Next, you will be prompted to enter a cooler type at a facility:
 - 'HGF' for Hawkeye
 - 'BBCS2' for Blue Buttes 2
 - 'BWCS' for Buffalo Wallow
 - 'BBCS' for Legacy BB

5. If BBBS2 or BWCS selected, follow prompts to save the generated analysis (**fully automated**)

If HGF or Legacy BB selected, follow additional steps below for exporting Operator Rounds data, required for computation

6. Saving Plots: plots must be **manually saved** (click the save icon at the bottom right of the plot window):



- If you forget to save a plot, open the workbook and delete the most recent run sheet, then re-run the program and re-submit pertinent parameters

AS OF 07/29/2025,

- BWCS 1-4 PI TAGS ARE INSERTED WITHIN PROGRAM—AWAITING DATA FROM TRANSMITTERS
- HGF FG 2&3 UNAVAILABLE
- HGF REFRIG 2 UNAVAILABLE

SoR Fin Fan Cooler Analysis Tool – Notes and Reminders

- Excel workbook name must be exact (case sensitive) and in the same folder as application

- There are 6 presently:

	BBCS2Coolers		7/17/2025 11:35 AM	Microsoft Excel W...	9 KB
	BBCSCoolers		7/11/2025 6:47 PM	Microsoft Excel W...	9 KB
	BWCSCoolers		7/11/2025 6:48 PM	Microsoft Excel W...	10 KB
	HGFFGCoolers		7/11/2025 6:50 PM	Microsoft Excel W...	10 KB
	HGFInletCoolers		7/11/2025 6:52 PM	Microsoft Excel W...	9 KB
	HGFRefrigCoolers		7/17/2025 11:33 AM	Microsoft Excel W...	9 KB
	SoR Fin Fan Analysis Tool		7/17/2025 11:46 AM	Application	50,100 KB

- (If using Operator Rounds from Webview AMS – Required for HGF and Legacy

BB) Access compressor discharge temps by connecting to Webview AMS at <https://www.webviewams.com/planning.aspx>. Here, you must export the Responses view.

- (If using Operator Rounds from Webview AMS – Required for HGF and Legacy

BB) Once logged in, from Planning, go to a date and select 'DONE' → Select a unit → 'Responses':

The screenshot shows the WebviewAMS Planning interface. The left sidebar contains navigation options like My Dashboard, Rounds/Inspections, Exceptions, Planning, Loc/Assets, Log Books, Documents, Reports, My Profile, Messages, Settings, and Sign Out. The main area displays a calendar for 7/23/2025. Below the calendar, a table lists responses for 'HAWKEYE: COMPRESSOR C1 (29)'. The table has columns for Label, Response, Alarm Config, Condition, Response Date/Time, Response By, and Comment. The responses include various temperature and pressure readings for different stages and inlets/outlets, all with a 'NORMAL' condition and a response time of 07/23/2025 08:32:00. The bottom of the screen shows '193 rows'.

- (If using Operator Rounds from Webview AMS – Required for HGF and Legacy

BB) Here, press the Excel sheet button located in the top left:

Label	Response	Alarm Config	Condition	Response Date/Time	Response By	Comment
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	AMBIENT TEMP	62	NORMAL	07/23/2025 08:26:47	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	COMPRESSOR STATUS	ONLINE	NORMAL	07/23/2025 08:26:47	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	SUCTION PRESSURE	11 PSI	NORMAL	07/23/2025 08:31:43	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	1ST STAGE PRESSURE	90 psig	NORMAL	07/23/2025 08:31:44	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	2ND STAGE PRESSURE	206 psig	NORMAL	07/23/2025 08:31:46	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	3RD STAGE PRESSURE	305 psig	NORMAL	07/23/2025 08:31:50	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	DISCHARGE PRESSURE	1178 PSI	NORMAL	07/23/2025 08:31:54	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	1ST STAGE INLET TEMP	55 degF	NORMAL	07/23/2025 08:31:59	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	2ND STAGE INLET TEMP	105 degF	NORMAL	07/23/2025 08:32:00	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	3RD STAGE INLET TEMP	106 degF	NORMAL	07/23/2025 08:32:03	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	4TH STAGE INLET TEMP	108 degF	NORMAL	07/23/2025 08:32:06	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	1ST STAGE #1 NORTH OUTLET TEMP	236 degF	NORMAL	07/23/2025 08:32:16	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	1ST STAGE #1 SOUTH OUTLET TEMP	234 degF	NORMAL	07/23/2025 08:32:17	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	1ST STAGE #3 NORTH OUTLET TEMP	226 degF	NORMAL	07/23/2025 08:32:28	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	1ST STAGE #3 SOUTH OUTLET TEMP	237 degF	NORMAL	07/23/2025 08:32:24	Auen, Randie	
HAWKEYE: COMPRESSOR 1-INLET GAS: HCS C1 (12294041)	1ST STAGE #5 NORTH OUTLET TEMP	228 degF	NORMAL	07/23/2025 08:32:47	Auen, Randie	

- (If using Operator Rounds from Webview AMS – Required for HGF) Once the sheet is downloaded, move it to the main folder along with the other sheets, and rename it according to how you are prompted to name it upon running the program:

You have selected HGF Inlet Gas Coolers

Verify that the operator rounds data is uploaded to this folder with the name 'InletRounds.xlsx' and press ENTER to continue...

Name	Status	Date modified	Type	Size
Original Calculators	✓	7/23/2025 6:55 AM	File folder	
BBCS2Coolers	✓	7/22/2025 11:35 AM	Microsoft Excel W...	17 KB
BBCSCoolers	✓	7/22/2025 10:28 AM	Microsoft Excel W...	16 KB
BWCSCoolers	✓	7/22/2025 11:32 AM	Microsoft Excel W...	17 KB
HGFFGCoolers	☁	7/22/2025 10:28 AM	Microsoft Excel W...	17 KB
HGFInletCoolers	✓	7/23/2025 11:18 AM	Microsoft Excel W...	17 KB
HGFRefrigCoolers	☁	7/22/2025 10:28 AM	Microsoft Excel W...	16 KB
InletRounds	✓	7/24/2025 8:49 AM	Microsoft Excel W...	19 KB
main	✓	7/22/2025 10:29 AM	Python Source File	168 KB
main.spec	☁	7/22/2025 10:29 AM	SPEC File	1 KB
requirements	☁	7/22/2025 10:29 AM	Text Document	2 KB
SoR Fin Fan Analysis Tool	🔄	7/23/2025 10:42 AM	Application	50,100 KB
SoR Fin Fan Cooler Analysis Tool User ...	✓	7/22/2025 10:29 AM	Adobe Acrobat D...	1,924 KB

Now, continue using the program as usual to obtain duties for these coolers

****NOTE:** Program does not work if any units are not running **that are not normally running** (not compatible with 'OOS' or 'STANDBY' for a unit that is usually on)

6. Understanding the excel sheet:

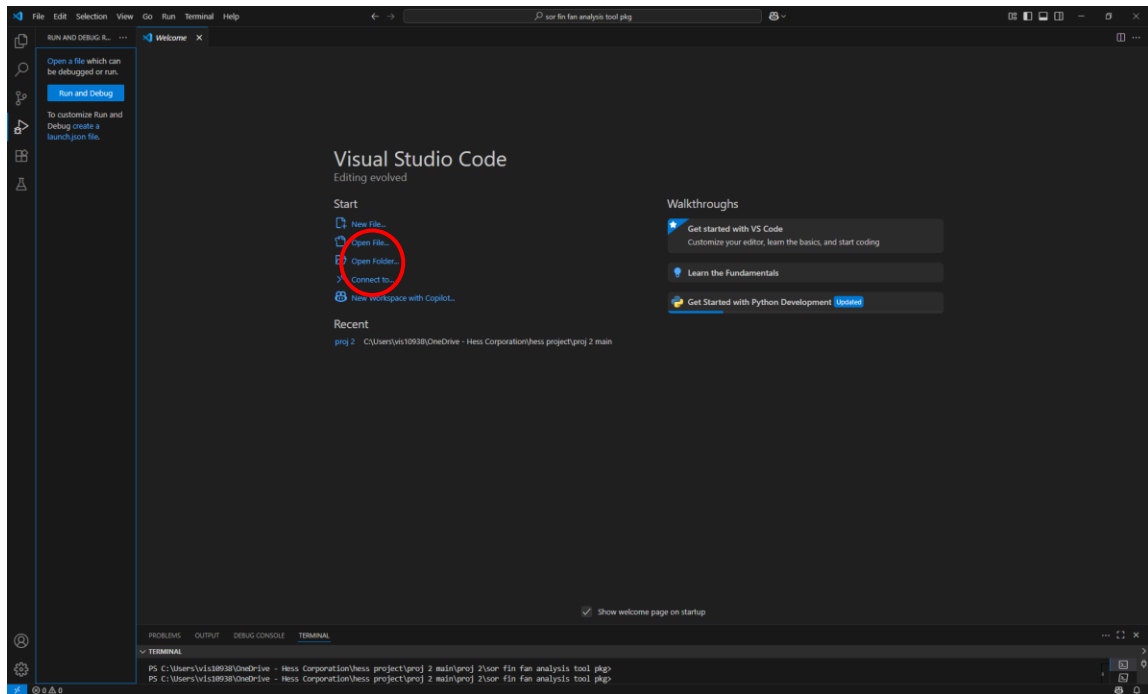
1. Whenever a run is executed, data is pulled and stored in one of the aforementioned excel workbooks as a new sheet:

[illegible]

The arrow points to the sheet and timestamp obtained for a single run, taken on 07/15/2025 at 22:34:50

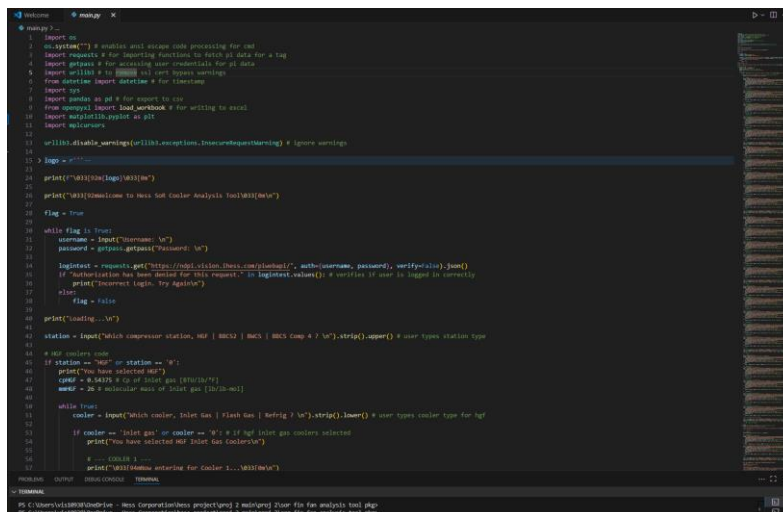
SoR Fin Fan Cooler Analysis Tool – Adding Units for Analysis for BWCS

1. Open python program in VS Code (free download at <https://code.visualstudio.com/>)
2. Next, open the application and select ‘Open Folder’:



Find the project folder containing the .py file (and all the excel workbooks, etc.)

3. Open ‘main.py’:

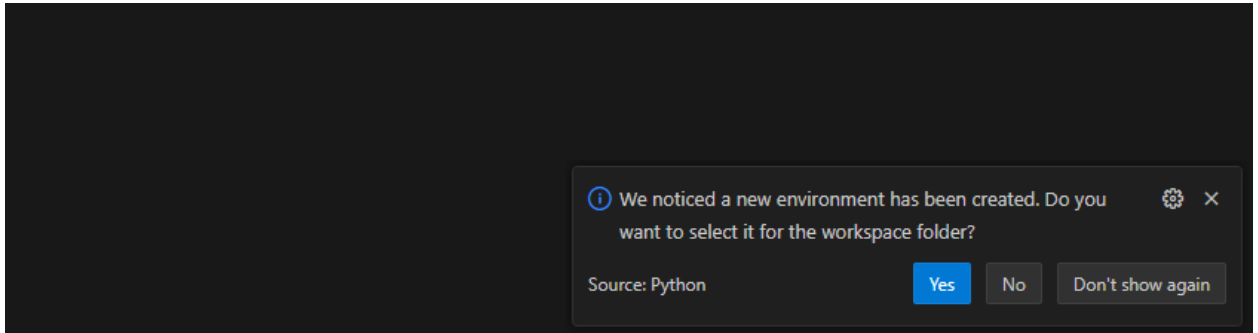


This is what the display will look like. The lower area is the terminal; the upper area is the bulk code. The terminal is where a few measures must be performed before beginning program manipulation.

4. In the terminal, execute the following command:

```
ysis tool pkg> python -m venv .venv  
ysis tool pkg> █
```

to create a virtual environment. This is necessary to run the dependency packages for various libraries in the code.



This message will appear in the bottom right corner. Select 'Yes'

5. Next in the terminal, execute the following command:

```
PS C:\Users\vis10938\OneDrive - Hess Corporation\hess project\proj 2 main\proj 2\sor fin fan analysis tool pkg> .venv\Scripts\activate  
(.venv) PS C:\Users\vis10938\OneDrive - Hess Corporation\hess project\proj 2 main\proj 2\sor fin fan analysis tool pkg> █
```

The second line is what should happen. You have entered the virtual environment to operate and execute the program.

6. Finally, install all the dependencies using the following line:

```
hess project\proj 2 main\proj 2\sor fin fan analysis tool pkg> .venv\Scripts\activate  
ration\hess project\proj 2 main\proj 2\sor fin fan analysis tool pkg> pip install -r requirements.txt █
```

Now, the main.py is in a workable state for further modification, testing, and packaging.

7. Now, go to 'BWCS coolers code' section:

```
main.py x imports Untitled 1
venv > main.py > ...
14
15 > logo = r'''...
23
24 print(f"\033[92m{logo}\033[0m")
25
26 print("\033[92mWelcome to Hess SoR Cooler Analysis Tool\033[0m\n")
27
28 flag = True
29
30 > while flag is True: ...
39
40 print("Loading...\n")
41
42 station = input("Which compressor station, HGF | BBSC2 | BWCS | BBSC Comp 4 ? \n").strip().upper() # user types station type
43
44 # HGF coolers code
45 > if station == "HGF" or station == '0': ...
624
625 # BBSC2 coolers code
626 > elif station == "BBSC2" or station == '1': ...
1497
1498 # BWCS coolers code
1499 > elif station == "BWCS" or station == '2': ...
1500
1501 # BBSC legacy cooler 4 code (ASSUMING COMP4 ONLY RUNNING COMPRESSOR)
1502 > elif station == "BBSC" or station == '3': ...
1503
1504 else: ...
1505
1506 # AFTER COLLECTING DATA
1507
1508 print("\033[92mAnalysis generation complete\033[0m")
1509
1510 while True:
1511     save = input("\nConfirm that you would like to save findings (Y/N).\n").strip().lower()
1512     if save == 'y': # if user selects to keep readings
1513         timestamp = datetime.today()
1514         if station == "BBSC2" or station == '1': ...
1515         elif station == "HGF" or station == '0': ...
1516         elif station == "BWCS" or station == '2': ...
1517         elif station == "BBSC" or station == '3': ...
1518     elif save == 'n': ...
1519     else: ...
1520
1521 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
1522
1523 PS C:\Users\vis10938\OneDrive - Hess Corporation\hess project\proj 2 main\proj 2>
1524 PS C:\Users\vis10938\OneDrive - Hess Corporation\hess project\proj 2 main\proj 2>
```

```
1498 # BWCS coolers code
1499 elif station == "BWCS" or station == '2':
1500     print("You have selected BBSC2")
1501     cpBW = 0.5436823 # Cp of inlet gas [BTU/lb/°F] -- NOTE: USING VALUE FOR BBSC2 UNTIL LAB TEST FOR BWCS - 07.09.25
1502     mmBW = 25.44 # molecular mass of inlet gas [lb/lb-mol] -- NOTE: USING VALUE FOR BBSC2 UNTIL LAB TEST FOR BWCS - 07.09.25
1503
1504     while True:
1505         print("Now generating cooler analysis of Coolers 1-4...\n")
1506
1507         # --- COOLER 1 ---
1508         volBW = requests.get("",
1509                               auth=(username, password), verify=False).json() # gets pi data (flowrate and timestamp) for Cooler 1
1510         if not isinstance(volBW["Value"], (int, float)): # if value has bad input
1511             print("\033[91mError: Bad transmitter input for Cooler 1 flow...\nDuty marked as 0 BTU/hr...\033[0m")
1512             mBW = 0
1513         else:
1514             mBW = volBW["Value"] * 1/397.3 * mmBW / 24 * 1000 # gets mass flow rate [1000-lb/hr]
1515
1516         # -- INTERCOOLER 1 --
1517         # gaining values for q computation
1518         T_in = requests.get("",
1519                              auth=(username, password), verify=False).json() # gets pi data for inlet temp
1520         if not isinstance(T_in["Value"], (int, float)): # if value has bad input
1521             print("\033[91mError: Bad transmitter input for Cooler 1 IC-1...\nDuty marked as 0 BTU/hr...\033[0m")
1522             q_BWC1_IC1 = 0
1523         else:
1524             T_out = requests.get("",
1525                                  auth=(username, password), verify=False).json() # gets pi data for outlet temp
1526             if not isinstance(T_out["Value"], (int, float)): # if value has bad input
1527                 print("\033[91mError: Bad transmitter input for Cooler 1 IC-1...\nDuty marked as 0 BTU/hr...\033[0m")
1528                 q_BWC1_IC1 = 0
1529             q_BWC1_IC1 = mBW * cpBW * (T_in["Value"] - T_out["Value"]) # heat rejection (1000-BTU/hr)
1530
1531         # -- INTERCOOLER 2 --
1532         # gaining values for q computation
1533         ...
```

This is where the bulk of the code is pertaining to the automated Pi tag data pull. The arrow points to what must be filled in in order to allow operation for this data automation point.

8. Go to Pi Web API at this link:

<https://ndpi.vision.ihess.com/piwebapi/dataservers/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA/points?maxCount=40000>

It will take a few minutes to load. This is a comprehensive list of all the links for our pi tags across 'NDPIDACOLL'

As you search for the correct Pi WebAPI link in the later sections, this 'maxCount' part will prove important—you can increase the amount of items it shows by increasing the number from 40000. This is done if in the initial 40,000 results, the Pi tag cannot be found.

9. Press 'Ctrl + F' to search the page, and enter the Pi tag to look for its API address.

There may be more than one result that it finds—ensure that the link pertains to the correct Pi tag (FIT vs. TIT vs. VIT etc.)

```
{
  "summaryData": "https://ndpi.vision.ihess.com/piwebapi/dataservers/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA/points?maxCount=40000",
  "value": "https://ndpi.vision.ihess.com/piwebapi/dataservers/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA/points?maxCount=40000",
  "endValue": "https://ndpi.vision.ihess.com/piwebapi/dataservers/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA/points?maxCount=40000"
},
{
  "id": "F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
  "id": "188818",
  "name": "USA,ND,HID.505.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageInletTemp.TIT792188A",
  "path": "\\\\NDPIDACOLL\\USA,ND,HID.505.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageInletTemp.TIT792188A",
  "descriptor": "TIT-TOP-95 - Blue Buttes Comp Station - Net Compressor1 - Suction Scrubber - Stage 1 - Compressor Cylinder - Temp Transmitter - Inlet - TIT-792188A - Scaling applied",
  "pointClass": "Classic",
  "pointType": "Float32",
  "digitalSetName": "",
  "engineeringUnits": "deg f",
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  "zero": -40.0,
  "step": false,
  "future": false,
  "displayDigits": -5,
  "links": {
    "self": "https://ndpi.vision.ihess.com/piwebapi/points/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "dataserver": "https://ndpi.vision.ihess.com/piwebapi/dataservers/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "attributes": "https://ndpi.vision.ihess.com/piwebapi/attributes/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "interpolateData": "https://ndpi.vision.ihess.com/piwebapi/interpolateData/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "recordedData": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "plotData": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "summaryData": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "value": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "endValue": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA"
  },
  "WebId": "F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
  "id": "188818",
  "name": "USA,ND,HID.505.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageIntercoolerTemp.TIT792188B",
  "path": "\\\\NDPIDACOLL\\USA,ND,HID.505.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageIntercoolerTemp.TIT792188B",
  "descriptor": "TIT-TOP-95 - Blue Buttes Comp Station - Net Compressor1 - Suction Scrubber - Stage 1 - Intercooler - Temp Transmitter - TIT-792188B - Scaling applied",
  "pointClass": "Classic",
  "pointType": "Float32",
  "digitalSetName": "",
  "engineeringUnits": "deg f",
  "span": 750.0,
  "zero": -40.0,
  "step": false,
  "future": false,
  "displayDigits": -5,
  "links": {
    "self": "https://ndpi.vision.ihess.com/piwebapi/points/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "dataserver": "https://ndpi.vision.ihess.com/piwebapi/dataservers/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "attributes": "https://ndpi.vision.ihess.com/piwebapi/attributes/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "interpolateData": "https://ndpi.vision.ihess.com/piwebapi/interpolateData/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "recordedData": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "plotData": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "summaryData": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "value": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "endValue": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA"
  },
  "WebId": "F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
  "id": "188818",
  "name": "USA,ND,HID.505.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageOutletTemp.TIT792188C",
  "path": "\\\\NDPIDACOLL\\USA,ND,HID.505.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageOutletTemp.TIT792188C",
  "descriptor": "TIT-TOP-95 - Blue Buttes Comp Station - Net Compressor1 - Suction Scrubber - Stage 1 - Compressor Cylinder - Temp Transmitter - Outlet - TIT-792188C - Scaling applied",
  "pointClass": "Classic",
  "pointType": "Float32",
  "digitalSetName": "",
  "engineeringUnits": "deg f",
  "span": 750.0,
  "zero": -40.0,
  "step": false,
  "future": false,
  "displayDigits": -5,
  "links": {
    "self": "https://ndpi.vision.ihess.com/piwebapi/points/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "dataserver": "https://ndpi.vision.ihess.com/piwebapi/dataservers/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "attributes": "https://ndpi.vision.ihess.com/piwebapi/attributes/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "interpolateData": "https://ndpi.vision.ihess.com/piwebapi/interpolateData/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "recordedData": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "plotData": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "summaryData": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "value": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA",
    "endValue": "https://ndpi.vision.ihess.com/piwebapi/streams/F1DSvZcIaSJkCZuCntY8tYTQTKRQSURBQ9MTA"
  }
}
```

10. Once the desired item is found, grab the link listed beside **“Value”**:. This is your API link that must go back into the program script

```

{
  "SummaryData": {
    "Name": "USA.ND.HID.565.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageInletTemp.TIT792108A",
    "Path": "\\\NDPI\DACOLL\USA.ND.HID.565.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageInletTemp.TIT792108A",
    "Descriptor": "TIT-792108 - Blue Buttes Comp Station - Inlet Compressor1 - Suction Scrubber - Stage 1 - Compressor Cylinder - Temp Transmitter - Inlet - TIT-792108A - Scaling applied",
    "PointClass": "classicc",
    "PointType": "Float32",
    "DigitalSetName": "",
    "EngineeringUnits": "deg f",
    "Span": 750.0,
    "Zero": -40.0,
    "Step": false,
    "Future": false,
    "DisplayDigits": 5,
    "Links": {
      "Self": "https://ndpi.vision.ihess.com/elemental/points/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "DataSet": "https://ndpi.vision.ihess.com/elemental/points/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "Attributes": "https://ndpi.vision.ihess.com/elemental/points/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "InterpolateData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "RecordData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "PlotData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "SummaryData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "Value": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "EndValue": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q"
    }
  },
  "UnitData": {
    "Name": "USA.ND.HID.565.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageIntercoolerTemp.TIT792113",
    "Path": "\\\NDPI\DACOLL\USA.ND.HID.565.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageIntercoolerTemp.TIT792113",
    "Descriptor": "TIT-792113 - Blue Buttes Comp Station - Inlet Compressor1 - Suction Scrubber - Stage 1 - Intercooler - Temp Transmitter - TIT-792113 - Scaling applied",
    "PointClass": "classicc",
    "PointType": "Float32",
    "DigitalSetName": "",
    "EngineeringUnits": "deg f",
    "Span": 750.0,
    "Zero": -40.0,
    "Step": false,
    "Future": false,
    "DisplayDigits": 5,
    "Links": {
      "Self": "https://ndpi.vision.ihess.com/elemental/points/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "DataSet": "https://ndpi.vision.ihess.com/elemental/points/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "Attributes": "https://ndpi.vision.ihess.com/elemental/points/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "InterpolateData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "RecordData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "PlotData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "SummaryData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "Value": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "EndValue": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q"
    }
  },
  "UnitData": {
    "Name": "USA.ND.HID.565.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageOutletTemp.TIT792108B",
    "Path": "\\\NDPI\DACOLL\USA.ND.HID.565.CompressorStation:BlueButtes2.GasProcess.GasCompression.InletRecip1.FirstStageOutletTemp.TIT792108B",
    "Descriptor": "TIT-792108 - Blue Buttes Comp Station - Inlet Compressor1 - Suction Scrubber - Stage 1 - Compressor Cylinder - Temp Transmitter - Outlet - TIT-792108B - Scaling applied",
    "PointClass": "classicc",
    "PointType": "Float32",
    "DigitalSetName": "",
    "EngineeringUnits": "deg f",
    "Span": 750.0,
    "Zero": -40.0,
    "Step": false,
    "Future": false,
    "DisplayDigits": 5,
    "Links": {
      "Self": "https://ndpi.vision.ihess.com/elemental/points/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "DataSet": "https://ndpi.vision.ihess.com/elemental/points/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "Attributes": "https://ndpi.vision.ihess.com/elemental/points/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "InterpolateData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "RecordData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "PlotData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "SummaryData": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "Value": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q",
      "EndValue": "https://ndpi.vision.ihess.com/elemental/streams/FIDP-VV-CIAS5KZC-NTYBTQTCICAT8QURBQ9HFTXVAREUThQUUEIL1MhY50T8IQAVYU9SU8RVEIPTJcCTFVQ1VAVETHSHQVNAQSORNTLdUBUMPTVBSRVNTSUOLK1OTEVAVGVSAKkZUJMAU8RBNV7T1kFVFRTVAVUE1LukchyHTA6Q"
    }
  }
}

```

Here is a matrix of all relevant Pi tags for BWCS:

Relevant Pi Tags												
Cooler #	T_in	IC-1	T_out	IC-2	T_in	T_out	IC-3	T_out	AC/IC-4	T_out	FIT	
1		822108	822112	822109	822113	822110	822114	822111	822115	822120		
2												
3												
4		822408	822412	822409	822413	822410	822414	822411	822415	822420		

11. Copy and paste the link and insert into the script in **between the quotations**:

```

main.py • import os Untitled-1
.venv > main.py > ...
25
26 > while flag is True:
27
28     print("Loading...\n")
29
30     station = input("Which compressor station, HGF | BCCS2 | BWCs | BCCS Comp 4 ? \n").strip().upper() # user types station type
31
32     # HGF coolers code
33     if station == "HGF" or station == "0":
34
35     # BCCS2 coolers code
36     elif station == "BCCS2" or station == "1":
37
38     # BWCs coolers code
39     elif station == "BWCs" or station == "2":
40         print("You have selected BWCs")
41         cpBW = 0.5436823 # cp of inlet gas [BTU/lb/°F] -- NOTE: USING VALUE FOR BCCS2 UNTIL LAB TEST FOR BWCs - 07.09.25
42         mmBW = 25.44 # molecular mass of inlet gas [lb/lb-mol] -- NOTE: USING VALUE FOR BCCS2 UNTIL LAB TEST FOR BWCs - 07.09.25
43
44         while True:
45             print("Now generating cooler analysis of Coolers 1-4...\n")
46
47             # --- COOLER 1 ---
48             volBW = requests.get("https://ndpi.vision.ihess.com/piwebapi/streams/F10PvVzCtasJsK2Ctnt8tYtQUsICAATkRQSURBQ89MTfXVU8EuTqkTUJELIMHuySDT8iQUkvU895U18RVEIPTjpcTFVfQ1VUVEVTHSHQVNVQK9D
49             auth=(username, password), verify=False).json() # gets pi data (flowrate and timestamp) for Cooler 1
50             if not isinstance(volBW["Value"], (int, float)): # if value has bad input
51                 print("\033[91mError: Bad transmitter input for Cooler 1 flow...\nDuty marked as 0 BTU/hr...\033[0m")
52                 mBW = 0
53             else:
54                 mBW = volBW["Value"] * 1/397.3 * mmBW / 24 * 1000 # gets mass flow rate [1000-lb/hr]
55
56             # --- INTERCOOLER 1 ---
57             # gaining values for q computation
58             T_in = requests.get("",
59             auth=(username, password), verify=False).json() # gets pi data for inlet temp
60             if not isinstance(T_in["Value"], (int, float)): # if value has bad input
61                 print("\033[91mError: Bad transmitter input for Cooler 1 IC-1...\nDuty marked as 0 BTU/hr...\033[0m")
62                 q_BWC1_IC1 = 0
63             else:
64                 T_out = requests.get("",
65                 auth=(username, password), verify=False).json() # gets pi data for outlet temp
66                 if not isinstance(T_out["Value"], (int, float)): # if value has bad input
67                     print("\033[91mError: Bad transmitter input for Cooler 1 IC-1...\nDuty marked as 0 BTU/hr...\033[0m")
68                     q_BWC1_IC1 = 0
69                 else:
70                     q_BWC1_IC1 = mBW * cpBW * (T_in["Value"] - T_out["Value"]) # heat rejection (1000-BTU/hr)
71
72             # --- INTERCOOLER 2 ---

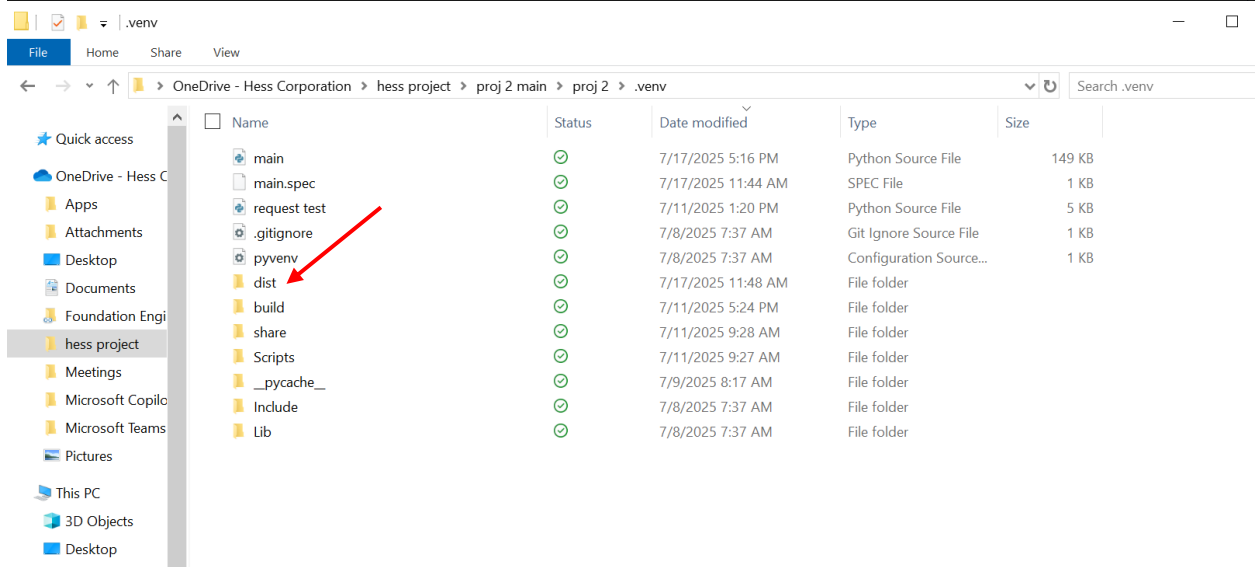
```

12. Repeat for all remaining inlet and outlet temps (currently, 2 per stage, 4 stages per cooler, 4 coolers for the station)
13. Then, use the line **'pyinstaller --onefile main.py'** in order to package the app. It will be found in a folder labeled 'dist':

```

PS C:\Users\vis10938\OneDrive - Hess Corporation\hess project\proj 2 main\proj 2> pyinstaller --onefile main.py

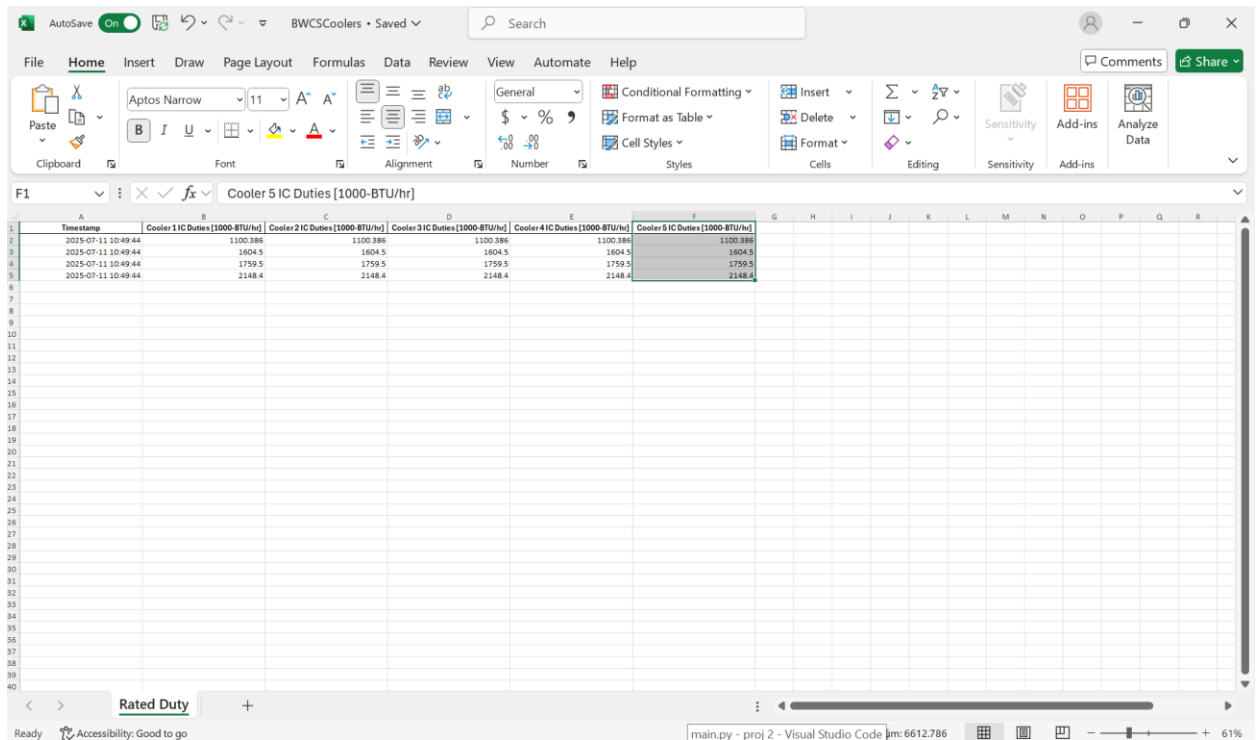
```



14. Finally, move the .exe file to ensure it is housed within the folder with all of the cooler excel workbooks. It will be titled **'main.py'**. Feel free to rename the file as desired. The updated tool will continue to integrate the existing data

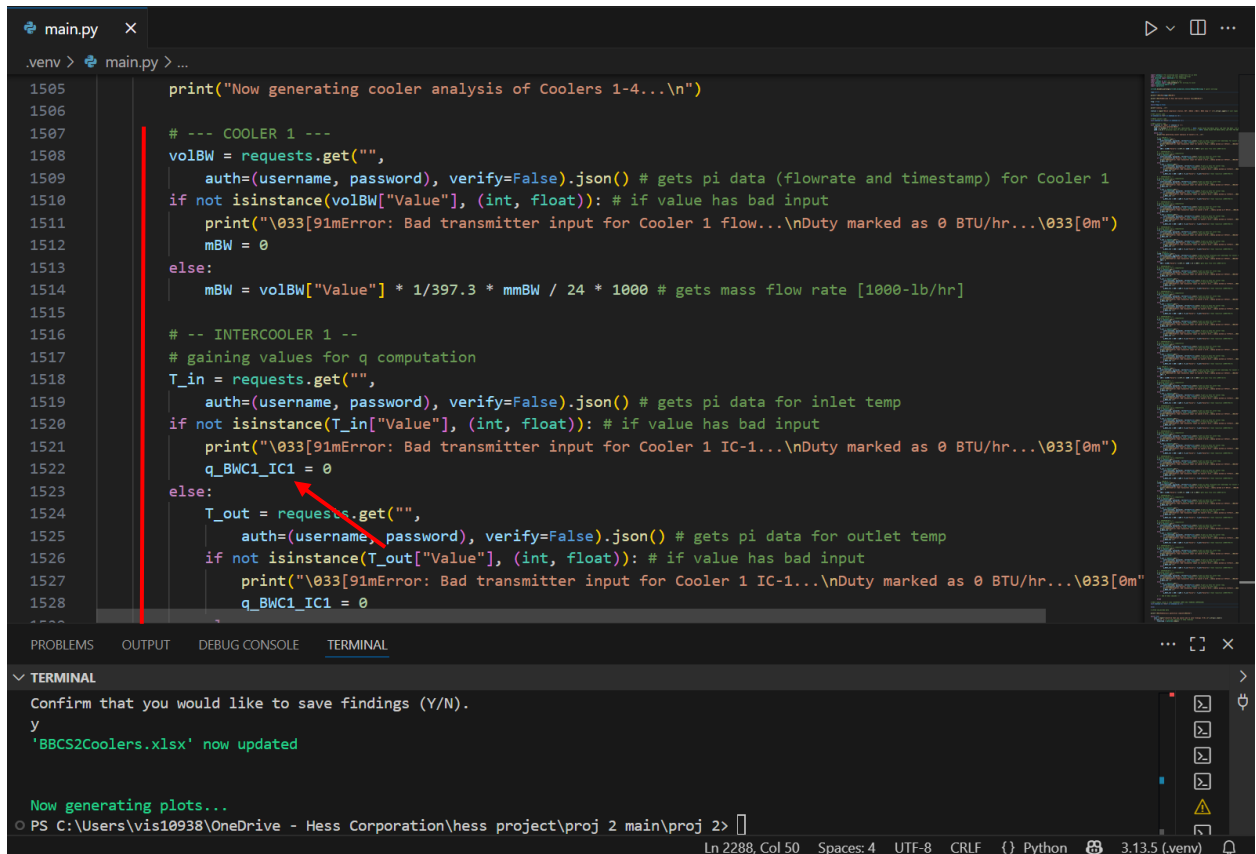
	BBCS2Coolers		7/17/2025 11:35 AM	Microsoft Excel W...	9 KB
	BBCSCoolers		7/11/2025 6:47 PM	Microsoft Excel W...	9 KB
	BWCSCoolers		7/11/2025 6:48 PM	Microsoft Excel W...	10 KB
	HGFFGCoolers		7/11/2025 6:50 PM	Microsoft Excel W...	10 KB
	HGFinletCoolers		7/11/2025 6:52 PM	Microsoft Excel W...	9 KB
	HGFRfrigCoolers		7/17/2025 11:33 AM	Microsoft Excel W...	9 KB
	SoR Fin Fan Analysis Tool		7/17/2025 11:46 AM	Application	50,100 KB

15. **(If new coolers are added)** An additional change must be made if new coolers are brought on: the excel workbook needs to be expanded in order to communicate with the program. A new **column** must be added to the first sheet in the workbook (even if it is a previous run—just substitute 0 for all the values). If an additional stage of cooling is added instead (such as EJW or TAW), a new **row** must be added.



Timestamp	Cooler 1 IC Duties [1000-BTU/hr]	Cooler 2 IC Duties [1000-BTU/hr]	Cooler 3 IC Duties [1000-BTU/hr]	Cooler 4 IC Duties [1000-BTU/hr]	Cooler 5 IC Duties [1000-BTU/hr]
2025-07-11 10:49:44	1100.386	1100.386	1100.386	1100.386	1100.386
2025-07-11 10:49:44	1604.5	1604.5	1604.5	1604.5	1604.5
2025-07-11 10:49:44	1759.5	1759.5	1759.5	1759.5	1759.5
2025-07-11 10:49:44	2148.4	2148.4	2148.4	2148.4	2148.4

16. **(If new coolers are added)** Next, the program must be updated in its visualization procedures to include the newly added data. First, duplicate the segments of the main loop that look like this, changing the end variable:



The screenshot shows a Python IDE with a file named `main.py`. The code is a script for generating cooler analysis. It includes comments and print statements for Cooler 1 and Intercooler 1. A red arrow points to the variable `q_BWC1_IC1` in the code. The terminal window at the bottom shows the following output:

```
Confirm that you would like to save findings (Y/N).
y
'BBCS2Coolers.xlsx' now updated

Now generating plots...
PS C:\Users\vis10938\OneDrive - Hess Corporation\hess project\proj 2 main\proj 2>
```

In the case of a 5th cooler to BWCS, once we copy and paste this template code at the bottom of the BWCS section, the variable must be changed to 'q_BWC5_IC1,' indicating the duty for 1st stage of cooler 5. This is repeated for stages 2-4 (and so on).

17. (If new coolers are added) Once the main code is updated, go to the section labeled 'BWCS plotting.' Here, the new variable(s) must be added to the data frame:

```
main.py 4 •
venv > main.py > ...
1875 while True:
1876     save = input("\nConfirm that you would like to save findings (Y/N).\n").strip().lower()
1877     if save == 'y': # if user selects to keep readings
1878         timestamp = datetime.today()
1879
1880         # bbcs2 plotting
1881         if station == "BBCS2" or station == '1':...
1882
1883         # hgf plotting
1884         elif station == 'HGF' or station == '0':...
1885
1886         # BWCS plotting
1887         elif station == "BWCS" or station == '2':
1888             BBCS2df = pd.DataFrame({
1889                 "Timestamp": timestamp,
1890                 "Cooler 1 IC Duties [1000-BTU/hr]": [q_BWC1_IC1, q_BWC1_IC2, q_BWC1_IC3, q_BWC1_IC4],
1891                 "Cooler 2 IC Duties [1000-BTU/hr]": [q_BWC2_IC1, q_BWC2_IC2, q_BWC2_IC3, q_BWC2_IC4],
1892                 "Cooler 3 IC Duties [1000-BTU/hr]": [q_BWC3_IC1, q_BWC3_IC2, q_BWC3_IC3, q_BWC3_IC4],
1893                 "Cooler 4 IC Duties [1000-BTU/hr]": [q_BWC4_IC1, q_BWC4_IC2, q_BWC4_IC3, q_BWC4_IC4],
1894                 "Cooler 5 IC Duties [1000-BTU/hr]": [q_BWC5_IC1, q_BWC5_IC2, q_BWC5_IC3, q_BWC5_IC4]
1895             })
1896
1897         # writing data to excel sheet
1898         sheetName = datetime.now().strftime("Run_%m%d%Y_%H%M%S") # labels sheet
1899
1900 PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL
```

(Alternate Case): Rather than adding a new cooler, if a new stage of cooling not listed (e.g. TAW) is to be added, a few more steps in this section will be necessary to be taken:

18. (Alternate Case) Perform the previous step but in a different manner (for adding another stage of cooling to pre-existing coolers):

```
main.py 4 •
venv > main.py > ...
1875 save = input("\nConfirm that you would like to save findings (Y/N).\n").strip().lower()
1876 if save == 'y': # if user selects to keep readings
1877     timestamp = datetime.today()
1878
1879     # bbcs2 plotting
1880     if station == "BBCS2" or station == '1':...
1881
1882     # hgf plotting
1883     elif station == 'HGF' or station == '0':...
1884
1885     # BWCS plotting
1886     elif station == "BWCS" or station == '2':
1887         BBCS2df = pd.DataFrame({
1888             "Timestamp": timestamp,
1889             "Cooler 1 IC Duties [1000-BTU/hr]": [q_BWC1_IC1, q_BWC1_IC2, q_BWC1_IC3, q_BWC1_IC4, q_BW1_TAW],
1890             "Cooler 2 IC Duties [1000-BTU/hr]": [q_BWC2_IC1, q_BWC2_IC2, q_BWC2_IC3, q_BWC2_IC4, q_BW2_TAW],
1891             "Cooler 3 IC Duties [1000-BTU/hr]": [q_BWC3_IC1, q_BWC3_IC2, q_BWC3_IC3, q_BWC3_IC4, q_BW3_TAW],
1892             "Cooler 4 IC Duties [1000-BTU/hr]": [q_BWC4_IC1, q_BWC4_IC2, q_BWC4_IC3, q_BWC4_IC4, q_BW4_TAW]
1893         })
1894
1895     # writing data to excel sheet
1896     sheetName = datetime.now().strftime("Run_%m%d%Y_%H%M%S") # labels sheet
1897
1898     workbook = load_workbook("BWCSCoolers.xlsx")
1899     writer = pd.ExcelWriter("BWCSCoolers.xlsx", engine="openpyxl", mode="a")
1900
1901 PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL
```

19. (Alternate Case) Add a data label tag in the 'labels' variable:

```
main.py
.venv > main.py > ...
2238     # Convert time column to datetime
2239     for name, df in xls.items():
2240         df[time_col] = pd.to_datetime(df[time_col])
2241
2242     cooler_cols = [col for col in existing_df.columns if col != time_col]
2243
2244     for col in cooler_cols:
2245         plt.figure(figsize=(8, 5))
2246         scatter_objects = [] # Store scatter plots for this figure
2247
2248         for sheet_name, df in xls.items():
2249             x_vals = df[time_col]
2250             y_vals = df[col]
2251
2252             # Scatter plot
2253             sc = plt.scatter(x_vals, y_vals, label=sheet_name)
2254             scatter_objects.append((sc, df, sheet_name, col))
2255
2256             labels = ["IC-1", "IC-2", "IC-3", "IC-4", "TAW"]
2257
2258             # Static labels
2259             for i in range(len(x_vals)):
2260                 plt.annotate(
2261                     labels[i],
```

And you are done. The plots will update with an added data point for the extra value(s).